

DUET: Exploring Event Visualizations on Timelines

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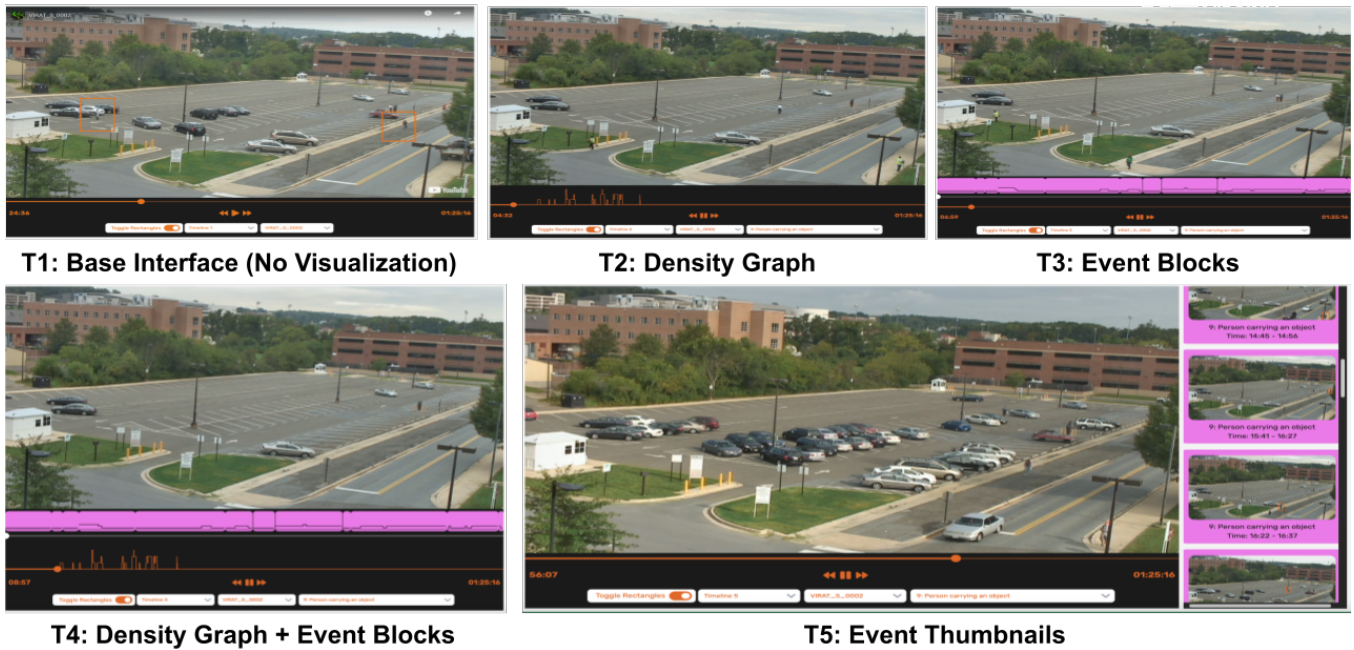


Figure 1: This is a preview of the five timeline visualizations that we developed. T1 is the base interface where the user only has basic video controls. T2 adds the density graph which helps the user temporally perceive where the events occur in the timeline. T3 adds event blocks which provide a space-efficient method of event visualization by removing the periods of downtime. T4 combines both T2 and T3 to create a visualization that offers both temporal perception of events and space-efficient event navigation. Lastly, T5 adds event thumbnails for a visualization that offers immediate visual feedback similar to other video players.

Abstract

Through innovations in sensors and computer vision, event-based temporal data collection has increased significantly in recent years. However, little, if any, research has focused on improving GUIs for these systems. Many applications use a linear timeline, leading to large downtime gaps when there are sparse events. In the meantime, linear timelines benefit from creating a natural sense of temporal perception. Our project seeks to find a balance between the natural temporal perception that linear timelines offer and improving the usability of visualizing event-based data using timelines. Through

the exploration of design alternatives, we aim to evaluate different types of timeline visualization through which users can leverage the temporal perception from linear timelines and quickly locate events of interest. We conducted two studies to evaluate our interfaces, first a formative assessment study with specialists and then an experimental study. Our findings inform the design of future event-based timeline visualizations.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Graph drawings*; *Visual analytics*; *User studies*; *Usability testing*.

Keywords

timeline, visualization, event-based, usability, data-processing

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1 Introduction

As humans continue to use technology more and more in their daily lives, collecting large amounts of event-based data in video footage has become easier than ever. Event-based data, or information related to specific events, including when, why, and how they occurred, can be collected digitally as people use their devices or through a combination of sensors and computer vision in different contexts (e.g., security cameras, pet cam, baby monitor camera, bird watchers, etc.) [18]. These types of event-based data with video can be used to identify patterns and trends in users' data or video footage. For example, the interaction trace in educational videos can be visualized as a density graph for other viewers to leverage the visualization and know what other learners frequently watch [8]. Additional visual cues, such as thumbnails, laid out on a timeline can be useful for users to quickly locate events of their interests [6]. However, lengthy videos, such as security camera footage, bring the new challenge of having to visualize sparse events in a much longer timeline (e.g., A doorbell camera recording with motion detection).

Although linear timelines with a density graph still offer effective temporal perception for extended videos since users can easily tell when events occur, scrubbing the timeline to locate a certain event will be challenging for users when there are sparse events in a lengthy video; users may have to scroll delicately and thoroughly as each move will correspond to a significant jump in time. Other applications have attempted to use a vertical list without the gaps [10], i.e., a list of short videos only when there's motion detected. Still, users have limited information to comprehend events and their sequential and temporal relationship at a glance (e.g., how long a suspect has been in a building) when presented with the list where time is not visualized.

Our project seeks to find a balance between the natural temporal perception that linear timelines offer and improving the usability of event-based data visualizations. We test this balance through the DUET (Distinct but United Event-based Timelines) project¹, various types of timeline interfaces to search for events in videos. The website has five different timeline visualizations for users to find events inside the videos.

Through evaluating the interfaces, we set out to answer the following research questions:

- (1) What difficulties do users face when searching for specific events in long timelines with sparse events?
- (2) How do different event timeline visualizations influence users' performance in locating events?

To gather preliminary thoughts and improve our interface before conducting an experimental study, we conducted a formative assessment study (N=5), where we interviewed those who regularly reviewed security camera footage. After implementing the improvements based on the findings from the interview study, we conducted

an evaluation study (N=72), where we recruited participants to complete two event-searching tasks per timeline visualization for each video and then a short exit questionnaire based on their thoughts about the timeline visualization. From the user study, we found significant differences in perceived usability, measured by SUS, between timelines. In contrast to the variance in their perception, we also noted that all types of event visualization, except the one with visualization, were comparably effective for users in finding specific events. From the results of our studies, we assert that the timeline visualizations can be used to inform the design of future event-based timeline visualizations.

2 Review of Literature**2.1 Use Cases for Event-based Data**

In a survey of existing literature focused on the processing and analysis of event-based data, researchers found event-based data used in domains such as digital preservation, journalism, social media, and more [19]. For example, some researchers used event-based data gathered through life logging as a method to aid people with dementia in maintaining their identities [15]. Another group of researchers used event-based data for sports-journalism to propose a general framework for event detection and summary generation in broadcast sports videos [9]. Aside from the previous examples, a different set of researchers proposed a procedure which showed time-space diagrams can be constructed using automatically collected high-resolution event-based traffic data [23]. Event-based data can be used to improve processes in many diverse domains.

2.2 Augmenting Event-based Navigation Interfaces

As previously mentioned, linear timelines are one of the most popular methods for navigating through events due to the natural sense of temporal perception that they offer [11]. Many applications take advantage of this benefit whether that be in the form of a progress slider in a video or a recollection of significant events in history.

One example of previous research focuses on improving the educational video navigation using user interaction data. The researcher created LectureScope which included a 2D timeline with a user interaction density graph, keyword search using an interactive transcript, video highlights using storyboards, a word cloud, and bookmarks. Their augmentations drew attention to significant moments and allowed users to have better navigation control [8]. Another example is BrowseLine, a timeline visualization for users to identify temporal patterns from their browsing history. The researchers implemented a 2D timeline with rows as hours, a horizontal list of visited websites in that hour grouped by domains, domain filtering, and metadata previews. The findings of this study show the temporal data from traditional browsing histories can be leveraged through visual pattern recognition to enhance a user's ability to recall past browsing history [6]. Aside from web browsing histories, some focused on personal history through LifeLines which presents a youth record on a single screen, providing direct access to medical information, and promoting critical alerts. The findings of this study emphasize the importance of creating and iterating on visualizations of personal data, especially life events that happen every day [12].

¹<https://duet Timelines.vercel.app/>

From these works, it is clear that traditional navigation methods, although effective, benefit significantly in usability from new augmentations. To set our work apart from existing literature, we aim to isolate and focus on the specific timeline augmentations to see which augmentations aid with searching for events in videos.

3 Implementation

After reviewing previous literature, we were still left seeking an improved understanding of three aspects:

- (1) What augmentations do users want in timeline visualizations to aid with searching for events?
- (2) What are the users' difficulties when searching for events using existing timeline visualizations?
- (3) Would adding different augmentations lead users to find the correct events when searching?

From these gaps in knowledge, we designed alternative designs to address these gaps and implement these options on the DUET website. When we started implementing the website, we wanted to ensure that we had a visualization that offered a great sense of temporal perception, a visualization that was space efficient when searching for events, and a visualization that provided visual information about the events. For content, we used VIRAT, an existing surveillance video footage dataset and its annotation information (e.g., timestamps, detected objects, their location, and activity type, such as 'a person carrying an object') to visualize events on the timeline [14].

3.1 Base Interface

The base interface is a plain video-player-like UI where a timeline is present for temporal navigation, but no events are visualized. Underneath the video player is a standard linear timeline with the elapsed time, basic video controls, and total duration similar to a commercial video platform such as YouTube [22]. The next feature is a toggle for showing or hiding the bounding rectangles since the bounding rectangles display detected objects on a video. From the pilot study, we found that users sometimes want to turn off this bounding box as it may occlude some information the user wants to see. The last features are dropdown menus for selecting a video or timeline type. We refer to the base interface as (Figure 1-T1).

3.2 Density Graph

We implemented a density graph above the existing linear timeline, which is commonly found in the literature as well as commercial apps (e.g., most replayed sections on YouTube videos) [8, 22]. To create the density graph, we parse the events in the current video, incremented the graph at the starting point of an event, and decremented the graph after the stopping point of it to create a 2D timeline to inform the user temporally when events are occurring [2, 13, 21]. After we created the density graph, we added filtering based on event types by adding a third dropdown menu with all of the different event types, allowing users to select one of the event types in the video. To separate the density graph, we refer to the time visualization with the density graph as (Figure 1-T2).

3.3 Event Blocks

One potential problem of the density graph is that if the events are sparse in time, each event will be visualized in a narrow space, which makes it difficult to navigate within the duration that the event happens. For example, an event that happens over 18 seconds will occupy 0.5% of the timeline for an hour-long video (3600 seconds). In this case, users will not have too much space on the screen (0.5%) to navigate within an event flexibly. To address this challenge, we created a linear timeline with event blocks [10]. In our implementation, an event block is a block generated based on segments of a density graph only when certain events occur [2, 6, 21]. To generate the event blocks, we repurposed the density graph and then grouped the events between start and end points into one block. We also allowed users to click on an event block to fast-forward to the first event that occurs [5]. We then placed all the event blocks horizontally above the linear timeline with no periods of downtime in between them. We then synced the event blocks to a new linear timeline which would follow along with when an event occurred in the original timeline. The new timeline dedicated to event blocks will be useful for users to navigate within each event block for a wider space than the density-graph timeline. To separate the event blocks, we refer to the time visualization with the event blocks as (Figure 1-T3).

3.4 Density Graph + Event Blocks

We created a combined version of the two timelines: density graph and event blocks. The main purpose of this combined version was for users to gain the benefits of both of the features, being able to see temporally when events occur in the timeline as well as have a space-efficient method of navigating through events. To separate the combined visualization, we refer to the time visualization with the density graph + event blocks as (Figure 1-T4).

3.5 Event Thumbnails

Lastly, we decided to implement one more timeline visualization which shows a list of the events on the right side of the video player. Each of the events in the lists was displayed with a thumbnail of the first frame of the event, the event type, and the start time of that event [2, 6, 21]. We also implemented a function where when a user clicks on one of the events, they are fast-forwarded to the start time of that event so they can watch that event happen. The main purpose of creating the event list with thumbnails was to provide users with a visual method of navigation [22]. To separate the event thumbnails, we refer to the time visualization with the event thumbnails as (Figure 1-T5).

4 Formative Assessment Study

4.1 Study Design and Methods

After implementing the timeline visualizations, we conducted interviews with users with occupations where they view security camera footage to discuss what difficulties they face [16]. The interviews were conducted remotely via Zoom. Each interview took approximately an hour to complete and each participant was compensated with a \$30 electronic gift card for their time. The first section mainly

focused on expanding on each participant’s responses to the sign-up survey shown in Appendix A. Then, we sent the participant a link to the DUET website for the tutorial. During the tutorial, we showed them the base interface, how to switch between the different types of timelines and videos, and a brief overview of each of the timeline visualizations that we implemented. After the tutorial, we had each participant attempt to complete a task that we created to represent a real-world scenario. The task involved counting the number of people that fit certain criteria. We had each participant switch to a video with a stop sign and count how many people who were waiting at the stop sign were carrying a suitcase. After the task, we asked each participant some further questions about our timeline visualizations, we answered any questions the participants asked, thanked them for their time, and let them know the timeline for compensation.

During the interviews, we interviewed two Police Officers from different departments in our local area. The first police officer we interviewed was a detective for their precinct. The second police officer we interviewed was a crime prevention officer for our university’s police department. After interviewing the police officers, we used Prolific to recruit people who have experience using security camera footage in various contexts. The first participant we recruited from Prolific has been a homeowner for twenty years. The second participant was a student who worked on a farm during her undergraduate studies as an undergraduate researcher. The third participant was a Security Specialist who works in an office for the water district of a government agency. After the interviews, we gathered the transcriptions from the five interviews and conducted a thematic analysis [3]. For the entire codebook we created, please refer to Appendix B.

4.2 Improvements

After organizing the information from the subcategories, we used the information to inform improvements to our current timeline visualizations.

4.2.1 Base Interface. For the improvements that we made to T1 which are shown in Figure 2, we started by adding a method to speed up and slow down the playback rate of the video. We used YouTube’s player to play our videos so we implemented speed-up and slow-down buttons which would increase and decrease the playback rate respectively. We also added an indicator underneath the linear timeline to report to the user the video’s current playback rate.



Figure 2: Base Interface

4.2.2 Density Graph + Event Blocks. As for the improvements made to T2-T4, which are shown in Figure 3, we changed the dropdown menu for the event types to a menu where all the event types are shown to the user at once, and then when the user selects an event type, a radio button is checked. We decided to change the dropdown menu because we wanted users to be able to see all of the event types present in the video immediately when the interface is loaded.

We added this new select menu to a permanent sidebar that is on the left side of the video player.

We also implemented a feature where when the timeline is within an event, the density graph at that event will be highlighted in white so that the user knows which event is currently happening. We implemented a similar feature for T3 where when an event is occurring in the event block, the event block will have a white outline around it to notify the user. After implementing these two features, we also made a change to T4 where when a user hovers over an event block, the event block has a green outline and the density graph for that event block is highlighted in green so that the user can tell which event block corresponds to which peak on the density graph before they click that event block.

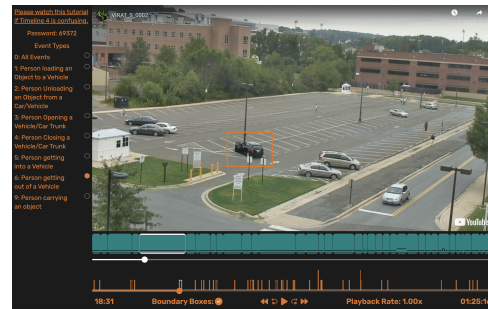


Figure 3: Density Graph + Event Blocks

4.2.3 Event Thumbnails. Lastly, for the improvements made to T5 which are shown in Figure 4, since the sidebar on the left was not being used by the menu for the event types, we made the list of thumbnails a horizontal list and placed it underneath the linear timeline and video controls. We also added a feature based on feedback from the pilot study where when a user clicks on one of the thumbnails, it will zoom into the image and allow the user to get a closer look at what action is happening in the thumbnail.



Figure 4: Event Thumbnails

5 Evaluation

5.1 Usability Study

After we made the improvements, we created a usability study to gain further data on what augmentations users wanted in timeline visualizations to aid them in searching for events. We decided to

use QuestionPro since we could convert the interview questions and tasks into a survey, which could be completed asynchronously by the participants. For our survey, we initially asked participants basic gender, age, and ethnicity questions followed by questions based on their previous usage of security camera video footage. After the introduction, we moved on to the tutorial section, where the participant was asked to watch a brief tutorial about how to use all five of the different timeline visualizations.

After finishing the tutorial, we created a QuestionPro block for each video. Each block begins by opening a link to DUET with a randomized timeline visualization. We then asked the participant to enter the password displayed on DUET, which was used to handle the timeline randomization. After entering the password, we asked the participants to solve two questions for that block. For each question, we also added an option for a user to choose if they could not find the answer within five minutes. After attempting to solve the two questions, we asked participants to select the timeline visualization that they used and judge the timeline visualization using the SUS questionnaire. After completing tasks, we asked each participant some exit questions before they completed the survey so that we could collect the user's thoughts about the different timeline visualizations. After the exit questions, we thanked the participants for their time, and let them know the timeline for compensation which was a 1 out of 10 chance to win a \$10 Amazon Gift Card.

5.2 Results

During the usability study, we recruited 72 participants through our University's mailing lists, local police stations, and other forms of online advertisements. To analyze the data collected from the survey and user study, we decided to focus on four different aspects, the number of questions the user got correct based on the timeline visualization they used, the participant's SUS Questionnaire score for the timeline visualization that they used, and the participant's personal ranking of the five timeline visualizations.

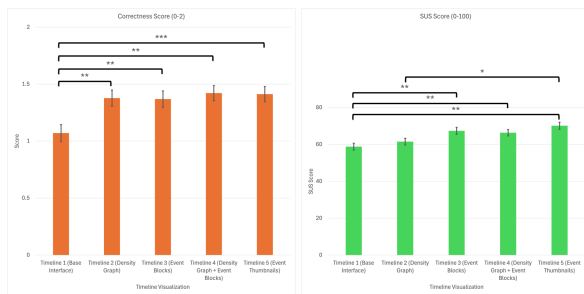


Figure 5: Correctness Bar Graph (Left) and SUS Bar Graph (Right)

5.2.1 Correctness Results. For the Correctness Results, we calculated a participant's correctness score based on how many questions the participant got correct for each timeline visualization. If a participant got none of the questions correct, their score was zero, if the participant got one question correct, their score was 1 and if the participant got both questions correct, their score was 2. After we got the scores from each participant for each of the five different

timeline visualizations, we calculated the mean and standard error for Correctness for each timeline visualization which is shown in Figure 5. Running a Kruskal Wallis test against the Correctness scores for all of the timeline visualizations, we see our Correctness scores are statistically significant ($p = 0.002$). Thus, we can conclude from the Correctness scores that the timeline visualization affects the participant's ability to find the correct events while searching for events in the video footage. For our post hoc analysis of the Correctness scores, we ran a Dunn's Multiple Comparison test which showed that all four other timeline visualizations led to higher correctness scores than T1.

5.2.2 SUS Questionnaire Results. For the SUS Questionnaire Results, we calculated the mean and standard error for the SUS Questionnaire scores for each of the timeline visualizations, which is shown in Figure 5. All five of the mean SUS Questionnaire scores are in the OK range when interpreting the SUS Questionnaire scores through adjectives [7]. Running a Kruskal Wallis test against the SUS Questionnaire scores for all timelines, we see our SUS Questionnaire scores are statistically significant ($p < 0.001$). Thus, we can conclude that our SUS Questionnaire scores show that the timeline visualization greatly affects the user's usability while searching for events in the video footage. For our post hoc analysis of the SUS Questionnaire scores, we ran a Dunn's Multiple Comparison test which showed that T3-T5 led to higher SUS Questionnaire scores than T1 and T5 led to higher SUS Questionnaire scores than T2.

6 Discussion and Future Work

Our study highlights some of the improvements that could be made to traditional timeline visualizations to help users navigate through long videos more seamlessly. We built upon existing concepts such as the traditional linear timeline, the density graph, and using thumbnails as well as implemented new methods for a user to navigate through events using the event blocks. Thus, we claim that the DUET augmentations can introduce more user-friendly methods of timeline navigation with events.

We also believe that DUET can be expanded to other domains aside from security camera footage. For example, the system could be used to search for different events that happen in videos which could help editors find scenes easily that they want to keep in the final video or cut out periods where not much is occurring [1]. Another application of this system could be to enhance the playback experience of game-play sessions. After competitive matches, players could use the visualizations to determine how they can improve their strategies for future matches or fans could use the visualizations to create highlight reels of their favorite players [4, 9]. Additionally moderators of large multiplayer servers could use the visualizations to determine instances of malpractice since they can detect when a large unusual spike of events occur [20]. Aside from the previous domains, one professional use case for the visualizations could be in cross-browser testing. The visualizations could be used to group together the automated clicks and other events that occur when performing cross-browser testing and then provide a better view of what each set of interactions is meant to mimic [17].

Of course, to expand our interface in the future, there would need to be more augmentations implemented into the system. One of the recommended augmentations from participants was showing a

preview when you hover over the scroll/playback bar, which would be very beneficial for providing more immediate visual feedback to the user. Another recommended augmentations from participants was the ability to pin or bookmark timestamps and events in the video, which would help a lot for users who want to quickly access certain events later on. We hope that any future timeline visualizations consider these augmentations.

Another future direction for the project could focus on how the density of events can impact the perceived usability of timeline visualizations. We hope that in the future, we can adapt the timeline visualization based on the event density since we could allow the user to zoom in to create a more sparse view of the events since it would only show the ones that are happening in that timeframe, additionally when the user zooms-out the timeline visualization would become dense again to show all of the events. We are confident that this improvement would help address another limitation of our study, which is that, in practice, the videos that are being reviewed can be much longer than the ones that we used in our study.

We assert that DUET's timeline visualizations and augmentations could be used to inform how to improve the usability of existing video players and timeline visualizations implemented in existing and future systems. The project and its associated data are available at this GitHub repository: <https://github.com/echo-lab/dual-timeline>.

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A Formative Assessment Study Signup Survey

- What is your current role at the organization and what responsibilities do you have in this role?
- How often do you view the security camera footage as a part of your work?
 - Every Day
 - Every Week
 - Every Month
 - Every Year
 - Other
- What are the typical events or incidents are the ones that you look for when you view the footage?
- What information do you typically use and are given as clues to spot the events? (time window, profile, or nothing?)
- Could you share the most challenging and time-consuming tasks that you had to do when reviewing camera footage? How long does it usually take for you to review the camera footage to find what you are looking for?

- What software does your organization use to view the footage that you collect from the security cameras?
- If there is software you use from other companies, are there any features that they share in common?
- Does your software have any special features for identifying events (i.e., computer vision for event identification)? If so, how does that help you find more quickly? When is it not helpful?
- Are there any current issues that you have with the current software that your organization uses to search for events across long periods of footage?

B Pilot Study Coding Session

B.1 Codes Not Related to Comments About Our Timeline Visualizations

- The context of reviewing camera footage
 - Crime-related
 - * C01:Reviewing camera footage is a primary task of the user:P01, P02, P03
 - * C02:Crime Reports may involve camera footage available:P01, P02
 - * C44:Home and community protection of neighbors involves security camera footage:P03
 - Long Time Frame
 - * C08:Case where the user does not know the timeframe of the incident of interest, video review can take a while:Video length can be up to two weeks:P01, P02
 - * Negative:C07:Case when the user does know the timeframe of the incident of interest, requests short 5-10 min video snippets, video review will be shorter:Significantly reduces video length needed for review:P01, P02
 - * C60:Some companies are obligated to keep the footage for a certain amount of time:P05
 - Video Software
 - * C16:Specialized software that users use to review footage:Security Desk, WatchGuard, Axon, ADT, Ring, GoPro Software, Axis:P02
 - * C17:Some users do not use specialized software to view the security footage, they just use the standard video player that their device comes with:P01, P02
 - Pre-Edited Footage
 - * C27:Loss Prevention Agents help the officers filter out the large videos for information they find useful:Liasion to law enforcement:P01
 - Alertness
 - * C45:Home security systems have an armed mode and unarmed mode:Can set it to be armed at away times:P03
 - * C62:For tracking users wearing an object, watch them for a little bit, if they are going offscreen, move on:P05
 - Research/Observation Related
 - * C51:New use case for security camera systems:Home monitoring, checking for wild animals eating plants, office security, checking for suspicious clients:P03, P04
 - * C54:Security camera system used to help aid in research/data collection:Have to reiew a lot more of the footage to make sure accurate data was being collected about the plants:P04
- What Users seek when reviewing camera footage
 - Reviewing not knowing what they will learn
 - * C04:Video Footage may not involve main incident but auxiliary information of suspect / vehicles:Footage may not show their face but can show two people breaking into a home:P01, P02, P03
 - * C08:Case where the user does not know the timeframe of the incident of interest, video review can take a while:Video length can be up to two weeks:P01, P02
 - Auxiliary information that are not the target incident
 - * C04:Video Footage may not involve main incident but auxiliary information of suspect / vehicles:Footage may not show their face but can show two people breaking into a home:P01, P02, P03
- Useful Video Reviewing Software Features
 - Motion Detection Indicators
 - * C10:Motion is visually annotated on the timeline / video or alerted to the user through notifications:Colored bar, red box:P01, P02, P03, P04, P05
 - * C46:Auxillary equipment to aid the security system:Mobile application with motion notifications and chime for motion alerts:P03
 - * C61:Camera is not recording all the time, usually only recording when it senses activity:P05
 - No Intelligence
 - * C11:Security Camera System does not have any advanced intelligent features:P02
 - * C17:Some users do not use specialized software to view the security footage, they just use the standard video player that their device comes with:P01, P02
 - Annotations/Bookmarks
 - * C18:Being able to mark/annotate timestamps in the footage for later review/storage and be able to navigate through them without temporal navigation:P01, P02
 - Multi-Window
 - * C30:Basic video players have multiple windows you can open:P01
 - * Negative:C31:Multiple views can be helpful but often views you can negate:P01
 - Playback speed
 - * C32:Some players have the option to fast forward through the footage:P01
 - Audio/Transcript
 - * C34:Feature to transcribe the audio in the security camera footage:Axon, transcribes audio and picks up on police warnings for body cams:P01
 - Timeline Zoom In / Out
 - * C42:Feature to zoom in or zoom out of the timeline bar using the scroll wheel:P01
 - Offline Video Saving
 - * C52:Able to download the data from the systems to store for offline use:P04
- Information given to users before reviewing footage
 - Visual Cues

- * C05:Users are given basic visual information about suspect before looking at the footage:biometric information, eye/hair color:P02
- * C28:Users are given basic information about shoplifting before looking at the footage:What was stolen, what time it occurred:P01
- * C06:Users are given basic information about vehicle appearance before looking at the footage:Vehicle make/model/color:P02
- * Negative:C64:Motion alerts can obstruct view of user and prevent identification of obscure visuals:P01
- Temporal Cues
 - * C63:Users are given temporal information about suspect's actions before looking at the footage:receipts, credit card information:P02
 - * C28:Users are given basic information about shoplifting before looking at the footage:What was stolen, what time it occurred:P01
 - * C63:Users are given temporal information about suspect's actions before looking at the footage:receipts, credit card information:P02
 - * C07:Case when the user does know the timeframe of the incident of interest, requests short 5-10 min video snippets, video review will be shorter:Significantly reduces video length needed for review:P01, P02
 - * Negative:C14:Eyewitnesses do not give any information about the timeframe of the incident of interest:P02
 - * Negative:C08:Case where the user does not know the timeframe of the incident of interest, video review can take a while:Video length can be up to two weeks:P01, P02
- Sequence of Events
 - * C33:Sequence of events in security footage:Kids scope out store, later return to rob store:P01
- Difficulties with reviewing camera footage
 - Challenges regarding multiple video sources
 - * C03:No standard across different businesses regarding video footage format, software ease of use:Disk/USB/File Player:P01, P02, P04
 - No / Limited intelligent functionality
 - * C09:AI features does not distinguish types of behaviour, only detects motion in view:P02, P03, P04
 - * C32:Some players have the option to fast forward through the footage:P01
 - * C53:Motion detection requires larger objects for some systems:Axis:P04
 - * C65:Some applications only have time based filters:P03
 - Massive amount of material to review
 - * C13:Cannot jump to specific instances, must continuously review video footage:P02, P04
 - * C29:Reviewing footage across large timeframes is basically impossible:P01
 - Massive amount of material to review even with intelligence
 - * C12:Videos that have lots of traffic, must review every instance of a person/vehicle/subject:Level of computer vision must be very detailed to filter out specifics:P02

- * C47:Some camera systems are prone to false positive results:Insects that fly into the view, Amazon deliveries:P03
- Incorrect Cues Given
 - * C15:Eyewitnesses do not give correct information about the timeframe of the incident of interest:P02
- Low Quality
 - * C35:Camera footage is not usually high quality:P01

B.2 Codes Related to Comments About Our Timeline Visualizations

- Navigability of current system
 - Event Filtering
 - * E19:Being able to have specific filters for types of events that occur in the videos to narrow down the search time using timeline 2:specific things like loading objects in unloading, uh, opening a vehicle, closing a vehicle, getting into a vehicle, getting out, carrying an object:P01, P02, P04
 - * E48:Current filtering method is more informative than existing home systems:P03
 - Reduce Search Time
 - * E20:Using the event blocks to click to instances of events and being able to let the events run to see if you found the right one to minimize searching:P02
 - * E38:Scenario for using the density graph feature:For a carjacking, do not need to watch the periods of downtime, can just watch the peaks:P01
 - Temporal Cues
 - * E21:Event Blocks and Density Graph give a better idea of where the event is and what is occurring during the time of the event:P03 initially disliked T4 but used it for task 1 and said this:P02, P03
 - User Interface
 - * E36:UI is user friendly and simplistic:Normal players have various features which make the UI very busy:P01
 - Toggling Features
 - * E39:Allow advanced features to be toggled on and off:P01, P03
- Timeline Type Preference Reasons
 - Finding Known Events
 - * E23:Prefer Thumbnails for finding a person carrying an object:P02
 - Immediate Visuals
 - * E56:Prefer Thumbnails for immediate visual of what is happening for an event:P04, P05
 - * E22:Prefer Thumbnails over the event blocks + Density graph:P02
 - Control over intelligent features
 - * E40:Prefer Density Graph over Event Blocks:P01: More confusion than the density graphs, density graphs give more control and less automation, make sure nothing is being cut out:P01
 - Confirmation
 - * E55:Prefer Thumbnails for checking if an event is mis-categorized or not::P04

- Suggested Video Reviewing Software Features
 - Video Traversal
 - * E41:Feature to grab the slider and move it along with the video:P01
 - Temporal Cues
 - * E25:Feature to change the opacity of event when a user has viewed it / Visual indicator so a user does not relick an already viewed event if they do not want to:P02
 - Fast Forwarding
 - * E37:Feature to increase the speed that the videos play at/Fast forward:P01, P03
 - Advanced Filters
 - * E26:Advanced Identification / Filtering:Look for a Red car, or Blue truck instead of just person / vehicle:P02
- Limitations with current Features
 - Thumbnails misguiding users
 - * E59:Thumbnails may lead to users assuming events are not the ones they are looking for:P04
 - Navigation
 - * E24:May make a mistake from relying on the image since one frame of a thumbnail is not enough for a whole event:P02
 - * E50:Thumbnails being only the first frame are not as helpful:P03
 - Massive amount of material to review
 - * E43:Need to be able to move timeline in small chunks to notice subtle changes:P01
 - * E58:Density graph may be overwhelming with lots of events:P04
 - * E49:Event Blocks and Density Graph give too much information/repetitive:P03
 - Temporal Cues
 - * Negative:E57:Do not prefer the event blocks because it is harder to see when events start and end:P04

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